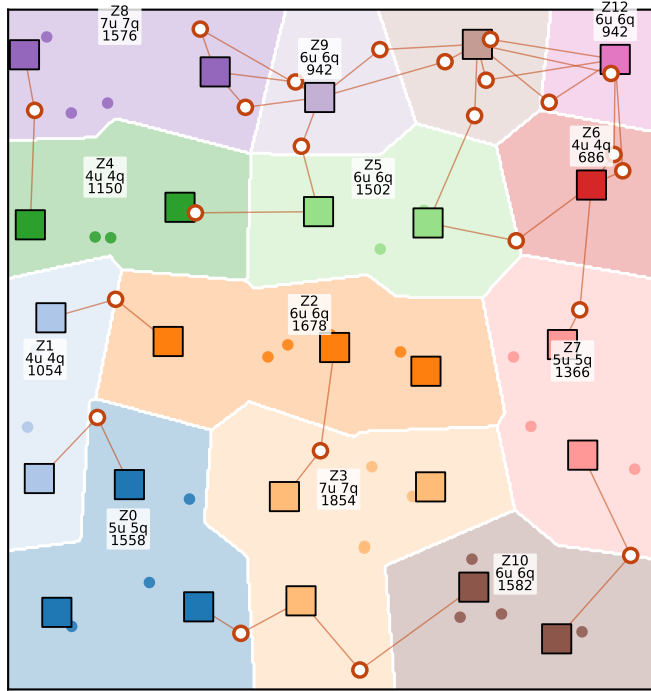
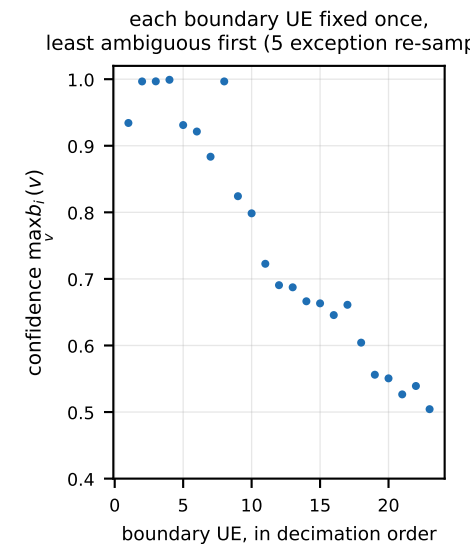
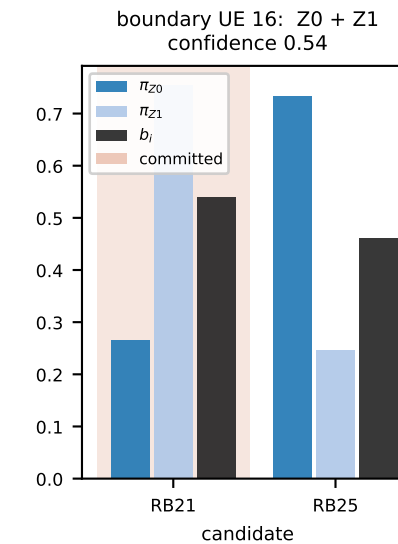
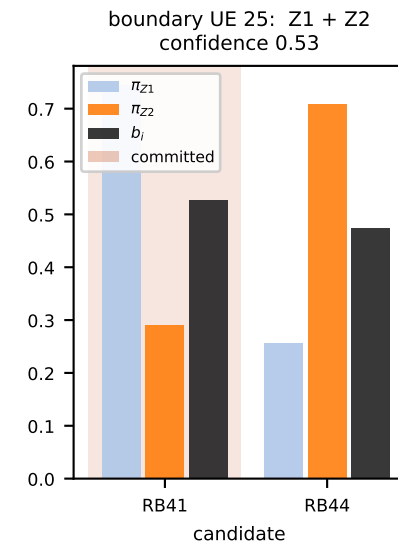
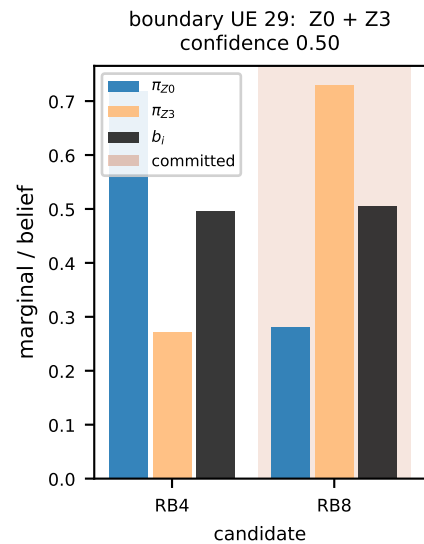
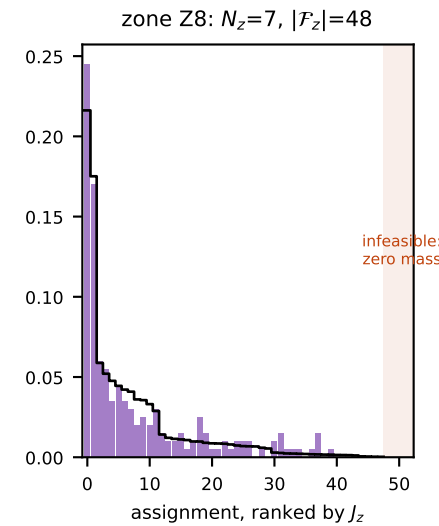
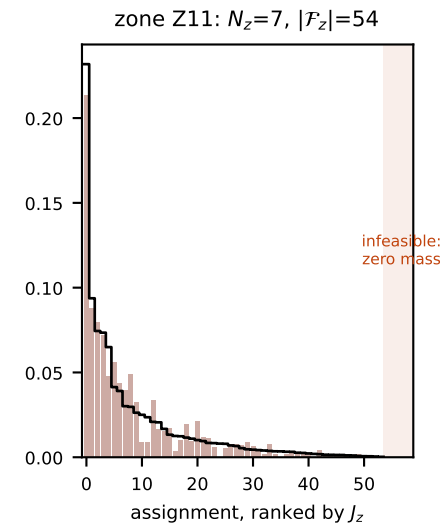
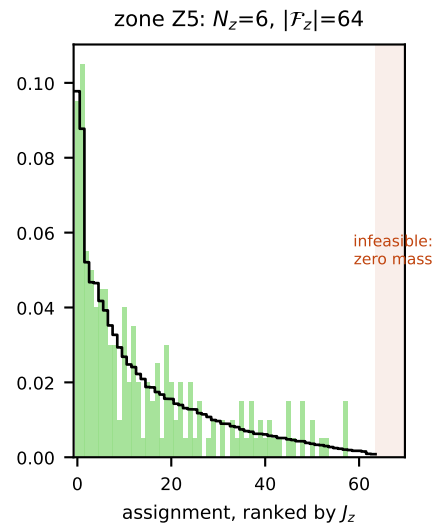
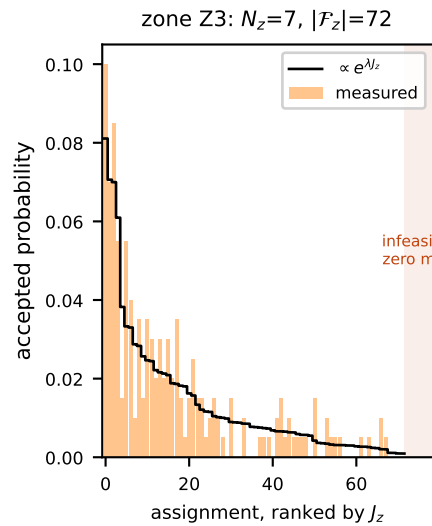


A region partitioned to the hardware budget, solved by zone-local sampling and classical boundary coordination

(a) partition under a 2000 two-qubit budget
25 APs, 50 UEs $\rightarrow$ 13 zones, $|\mathcal{B}|=23$ boundary UEs (46%)



shading: zone territory (nearest AP)
 AP (colour = zone) interior UE
 boundary UE UE-to-candidate-AP link



$g=5$, $seed=0$ (median of 8 seeds by utility ratio); $\beta=1.5$, $K_z=200$. Utility 297.2 of the centralized strict optimum 299.1 = 99.4%; the assignment is strictly feasible (True). Zone labels in (a): UEs, state qubits, two-qubit gates. Zone laws are exact: the accepted branch of the circuit matches $\exp(\lambda J_z)$ on $\mathcal{F}_z$ to statevector precision (max TVD $2.7e-15$ over the zones checked in `haiq_certify.py`), so the sampler used here is the same law the circuit produces.